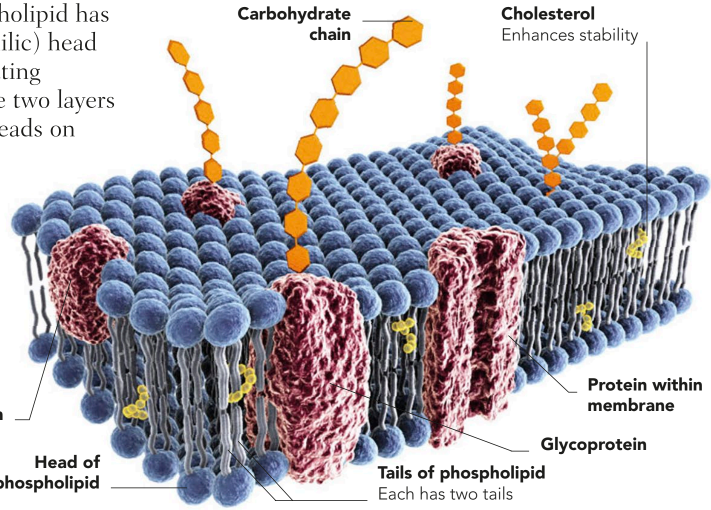
CELL MEMBRANE

Several features allow the membrane to fulfil its dual functions of protecting the cell's contents and permitting movement of materials into and out of the cell. The primary component of this membrane is a double layer of phospholipid molecules. Each phospholipid has a water-loving (hydrophilic) head group and two water-hating (hydrophobic) tails. The two layers are arranged with the heads on the outside and inside of the cell membrane, and the tails in between. The phospholipids are interspersed with protein molecules and carbohydrate chains that allow the cell to be recognized by other body cells.

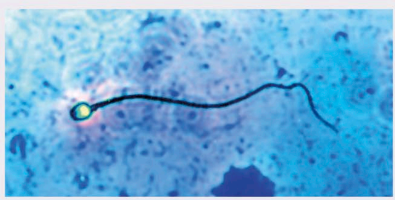


PERMEABLE BI-LAYER

The typical cell membrane is characterized by a double layer of phospholipids with embedded proteins.

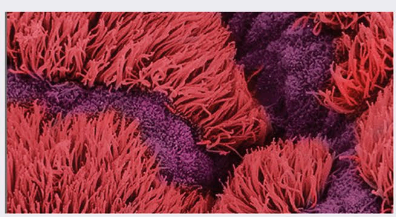
SURFACE ORGANELLES

Some cells in the body have specialized structures projecting from their surface. Cells lining the small intestine have small, finger-like projections called microvilli, which increase the surface area for absorption of nutrients. Some cells in the female reproductive tract have small, hair-like cilia that wave to move the ovum along the oviduct; similar ciliated cells in the respiratory tract move small particles out of the airways. The sperm is unique in the human body in having a long, whip-like flagellum, used for propulsion.



SPERM

The thin tail (flagellum) that extends from a human sperm cell is used like a propeller to help the sperm swim up the female genital tract.



CILIATED CELLS

Some of the cells lining the fallopian tubes have hair-like cilia (coloured pink in this micrograph) that brush an egg along towards the uterus.